Undergraduate Catalog

Computer Science (B.S.)

College

College of Engineering and Computer Science

Department

Department of Computer Science

Program Type

Major

Program Website

Computer Science B.S.

Program Contact Information

College of Engineering & Computer Science
Department of Computer Science

L3 Harris Corporation Engineering Center (HEC), 246
407-823-2341

CSundergrad@ucf.edu

***Associate Professor & Faculty Undergraduate Program Coordinator**: Dr. Mark Heinrich*
heinrich@ucf.edu

***Academic Program Coordinator**: Jenny Shen*
Jenny.Shen@ucf.edu

Is this program available 100% online?

No

Admission Requirements

None

Program Description

The computer science field addresses some of the most interesting and challenging problems of society. Computer scientists work at the forefront of technological progress and expanding the human intellect. They study phenomena connected with computers and computation including software (algorithms, data structures), and many diverse applications of computational thinking and techniques. Focus areas include robotics, artificial intelligence, cyber security, data analytics, simulation, game and other app design, data mining and more.

Computer Science emphasizes the mathematical and theoretical foundations of computing, rather than teaching specific technologies. A tremendous number of opportunities in a variety of fields awaits computer science graduates, including robotics, computer gaming, virtual reality, computer vision, media convergence, digital, evolutionary computing, computer architecture and so much more.

CECS Progress Policy

- Students in the Computer Science major are expected to make consistent good progress toward their degrees to remain enrolled in, or eligible for, any major, minor, or certificate in the College of Engineering & Computer Science (CECS) or the College of Optics and Photonics (COP). To support this, CECS and COP enforce a Progress Policy with two components: the low grade component and the repeat attempt component.
    - **Low Grade Component**: Any student majoring in Computer Science who has accumulated 7 or more unsuccessful attempts (i.e., grades below "C" (2.0), U grades, and withdrawals, including in 0-credit courses) over all courses taken at UCF will be placed on Lack of Progress Probation and remain on Lack of Progress Probation as long as the student is enrolled in a CECS or COP major. If a student on Lack of Progress Probation has a tenth unsuccessful attempt over all courses taken at UCF, the student will be Excluded from all CECS and COP majors, minors, and certificates.
    - **Repeat Attempt Component**: Any student majoring in Computer Science who repeats any UCF course and does not earn a grade of "C" (2.0) or better on the second attempt will be placed on Lack of Progress Probation and remain on Lack of Progress Probation as long as the student is enrolled in a CECS or COP major. If a student on Lack of Progress Probation does not receive a grade of "C" (2.0) or better by the third attempt in the same UCF course, the student will be Excluded from all CECS and COP majors, minors, and certificates.
- A student who has been excluded from CECS and COP majors cannot declare any CECS or COP minors or certificates unless as required by the student's degree program. A student who has been excluded from CECS and COP majors may seek readmission to a major in CECS or COP after earning a bachelor's degree or relevant associate of science degree, either from another college at UCF or from another institution. To be eligible for readmission to CECS or COP, the student must not have accumulated any additional low grades in subsequent coursework used to complete the earned degree. An excluded student who meets these criteria is not guaranteed readmission to a CECS or COP major. Any student who is readmitted to the Computer Science major will be subject to all probation conditions that applied at the time of exclusion.

General Education (GEP)

0

Total Credits

- Students should closely study the requirements of the UCF General Education Program (GEP) and the courses required for the major to minimize excess hours. Students transferring to UCF from within the Florida College System or State University System are encouraged to complete GEP requirements before transferring.

Communication Foundation 6 Total Credits

- Complete the following:
  - ENC1101 - Composition I (3)
  - ENC1102 - Composition II (3)

Humanities Foundation 6 Total Credits

- Complete at least 2 of the following:
  - HUM2020 - Encountering the Humanities (3)
  - MUL2010 - Enjoyment of Music (3)
  - PHI2010 - Introduction to Philosophy (3)
  - THE2000 - Theatre Survey (3)
  - ARH2050 - History of Western Art I (3)
  - ARH2051 - History of Western Art II (3)
  - EUH2000 - Western Civilization I (3)
  - EUH2001 - Western Civilization II (3)
  - FIL1000 - Cinema Survey (3)
  - FIL2030 - History of Motion Pictures (3)
  - HUM2210 - Studies in Culture: Ancient to 17th Century (3)
  - HUM2230 - Studies in Culture: 17th Century to Present (3)
  - LIT2110 - World Literature I (3)
  - LIT2120 - World Literature II (3)
  - MUH2017 - Survey of Rock Music (3)
  - MUH2019 - American Popular Music, 1840s-present (3)
  - MUH2111 - Survey of Music History: Antiquity to 1750 (3)
  - MUL2016 - Evolution of Jazz (3)
  - MUL2720 - Music of the World (3)
  - REL2300 - World Religions (3)
  - WOH2012 - World Civilization I (3)
  - WOH2022 - World Civilization II (3)

Mathematics Foundation 8 Total Credits

- The following courses are required for the major and can be used to satisfy the Mathematics Foundation.

8 Total Credits

- Complete the following:
  - MAC2311C - Calculus with Analytic Geometry I (4)
  - MAC2312 - Calculus with Analytic Geometry II (4)

Social Sciences Foundation

6
Total Credits

- Students must complete at least two courses from this foundation. It is recommended that students complete one of the following courses to satisfy Florida Civic Literacy Requirements (more information on this requirement can be found in the catalog under "Policies"): AMH 2010, AMH 2020, POS 2041

6
Total Credits

- Complete at least 2 of the following:
  - AMH2010 - U.S. History: 1492-1877 (3)
  - AMH2020 - U.S. History: 1877-Present (3)
  - POS2041 - American National Government (3)
  - ANT2000 - General Anthropology (3)
  - ECO2013 - Principles of Macroeconomics (3)
  - PSY2012 - General Psychology (3)
  - ECO2023 - Principles of Microeconomics (3)
  - HSA2117 - Civic Engagement in the US Healthcare System (3)
  - SYG2000 - Introduction to Sociology (3)

Natural Sciences Foundation

8
Total Credits

- The following courses are required for the major and can be used to satisfy the Natural Sciences Foundation.

8
Total Credits

- Complete all of the following
  - Earn at least 4 credits from the following course sets:
    - PHY 2048 and 2048L
      - PHY2048 - General Physics Using Calculus I (3)
      - PHY2048L - General Physics Using Calculus I Laboratory (1)
    - PHY 2048C
      - PHY2048C - General Physics Using Calculus I Studio (4)
  - Earn at least 4 credits from the following course sets:
    - PHY 2049 and 2049L
      - PHY2049 - General Physics Using Calculus II (3)
      - PHY2049L - General Physics Using Calculus II Laboratory (1)
    - PHY 2049C
      - PHY2049C - General Physics Using Calculus II Studio (4)

Additional GEP Courses

7
Total Credits

- Students must choose two additional courses from any of the five foundations to satisfy this requirement. The following courses are required for the major and can be used to satisfy this requirement.

7
Total Credits

- Complete all of the following
  - Complete the following:
    - STA2023 - Statistical Methods I (3)
  - Complete at least 1 of the following:
    - BSC2010C - Biology I (4)
    - CHM2045C - Chemistry Fundamentals I (4)

Grand Total Credits: 41

Common Program Prerequisites (CPP)

6
Total Credits

- Complete all of the following
  - A "C" (2.0) or better is required for all courses in this section.
  - Several CPP courses are met by courses specifically identified in the GEP: MAC 2311C, MAC 2312, PHY 2048C (or PHY 2048 and PHY 2048L), PHY 2049C (or PHY 2049 and PHY 2049L), and BSC 2010C or CHM 2045C.

Programming 3
Total Credits

- Earn a minimum grade of C in each of the following:
  - COP3223C - Introduction to Programming with C (3)

Mathematics 0
Total Credits

- Earn a minimum grade of C in each of the following: MAC 2311C - Calculus with Analytic Geometry I (4) and MAC 2312 - Calculus with Analytic Geometry II (4)

Physics I 0
Total Credits

- Complete 1 of the following
  - Earn a minimum grade of C in each of the following: PHY 2048 - General Physics Using Calculus I (3) and PHY 2048L - General Physics Using Calculus I Laboratory (1)
  - Earn a minimum grade of C in each of the following: PHY 2048C - General Physics Using Calculus I Studio (4)

Physics II 0 Total Credits

- Complete 1 of the following
  - Earn a minimum grade of C in each of the following: PHY 2049 - General Physics Using Calculus II (3) and PHY 2049L - General Physics Using Calculus II Laboratory (1)
  - Earn a minimum grade of C in each of the following: PHY 2049C - General Physics Using Calculus II Studio (4)

Science 3 Total Credits

- Complete all of the following
  - Earn at least 3 credits from the following types of courses:
    Earn a minimum grade of C in at least 2 of the following: BSC2010C - Biology I (4), BSC2011C - Biology II (4), CHM2045C - Chemistry Fundamentals I (4), CHM2046 - Chemistry Fundamentals II (3), and PHY3101 - General Physics Using Calculus III (3).
  - Students should ensure they complete at least 2 Science courses as listed above. Note that either BSC 2010C or CHM 2045C should be completed as part of the GEP, so for many students, only one additional Science course from this list is needed.

Grand Total Credits: 6

Degree Requirements

Basic Core Requirements 33 Total Credits

- A "C" (2.0) or better is required for all courses in Basic Core Requirements, Advanced Core Requirements, and Restricted Electives. STA 2023 is specifically identified in the GEP.

33 Total Credits

- Complete all of the following
  - Earn a minimum grade of C in each of the following:
    - COP3330 - Object Oriented Programming (3)
    - COT3100C - Introduction to Discrete Structures (3)
    - CDA3103C - Computer Logic and Organization (3)
    - CIS3360 - Security in Computing (3)
    - COP3502C - Computer Science I (3)
    - COP3503C - Computer Science II (3)
    - COP3402 - Systems Software (3)
    - COT4210 - Discrete Structures II (3)
    - COP4331C - Processes for Object-Oriented Software Development (3)
  - Complete at least 1 of the following:
    - CAP4611 - Algorithms for Machine Learning (3)

- CAP4630 - Artificial Intelligence (3)
- CAP4453 - Robot Vision (3)
- CAP4641 - Natural Language Processing (3)

- Earn a minimum grade of S in each of the following:
  - COT3960 - Foundation Exam
- Complete at least 1 of the following:
  - ENC3241 - Writing for the Technical Professional (3)
  - ENC3250 - Professional Writing (3)
- Earn a minimum grade of C in each of the following: STA 2023 - Statistical Methods I (3)

Advanced Core Requirements 24 Total Credits

- A "C" (2.0) or better is required for all courses in Basic Core Requirements, Advanced Core Requirements, and Restricted Electives. A minimum 2.500 GPA is required for courses in this section.

Technical Electives 18 Total Credits

- Complete all of the following
  - Earn at least 18 credits from the following types of courses:
    Any 4000-5000 level Computer Science course that is not a required undergraduate course will count as a technical elective.
  - Only ONE of the following six Information Technology courses may count as a Computer Science technical elective: CAP4102, CIS4004, CIS4524, CNT4603, CNT4703, and CNT4714.
  - At most 3 hours of independent study or research allowed. No cooperative education credits are allowed.
  - It is recommended that students take at least 3 classes from a primary area to gain an in-depth knowledge in that area of Computer Science. Refer to "Program Details" for more information on the primary areas.

Capstone Requirements 6 Total Credits

- Earn a minimum grade of C in each of the following:
  - COP4934 - Senior Design I (3)
  - COP4935 - Senior Design II (3)

Restricted Electives 9 Total Credits

- A "C" (2.0) or better is required for all courses in Basic Core Requirements, Advanced Core Requirements, and Restricted Electives.

Math/Statistics 3 Total Credits

- Earn at least 3 credits from the following types of courses:

MAC2313, MAP2302, MAS3105, MAS3106, or any 4000-5000 level Math/Statistics course with a MAA, MAD, MAS, or STA prefix, except independent study hours, internship, or cooperative education hours.

Flex STEM 6

Total Credits

- Earn at least 6 credits from the following types of courses:
  Any 2000-level or above course with the following prefixes: CS/IT (CAP, CGS, CDA, CIS, CNT, COP, COT), Engineering (BME, CCE, CEG, CES, CGN, CWR, EAS, EEE, EEL, EGN, EGS, EIN, EMA, EML, ENV, ESI, TTE), Mathematics (MAA, MAC, MAD, MAP, MAS, MTG, STA), Science (BCH, BOT, BSC, CHM, CHS, MCB, PCB, PHY, PHZ, ZOO)

Grand Total Credits: 66

Program Details

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4000 Central Florida Blvd. Orlando, Florida, 32816 | 407.823.2000

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